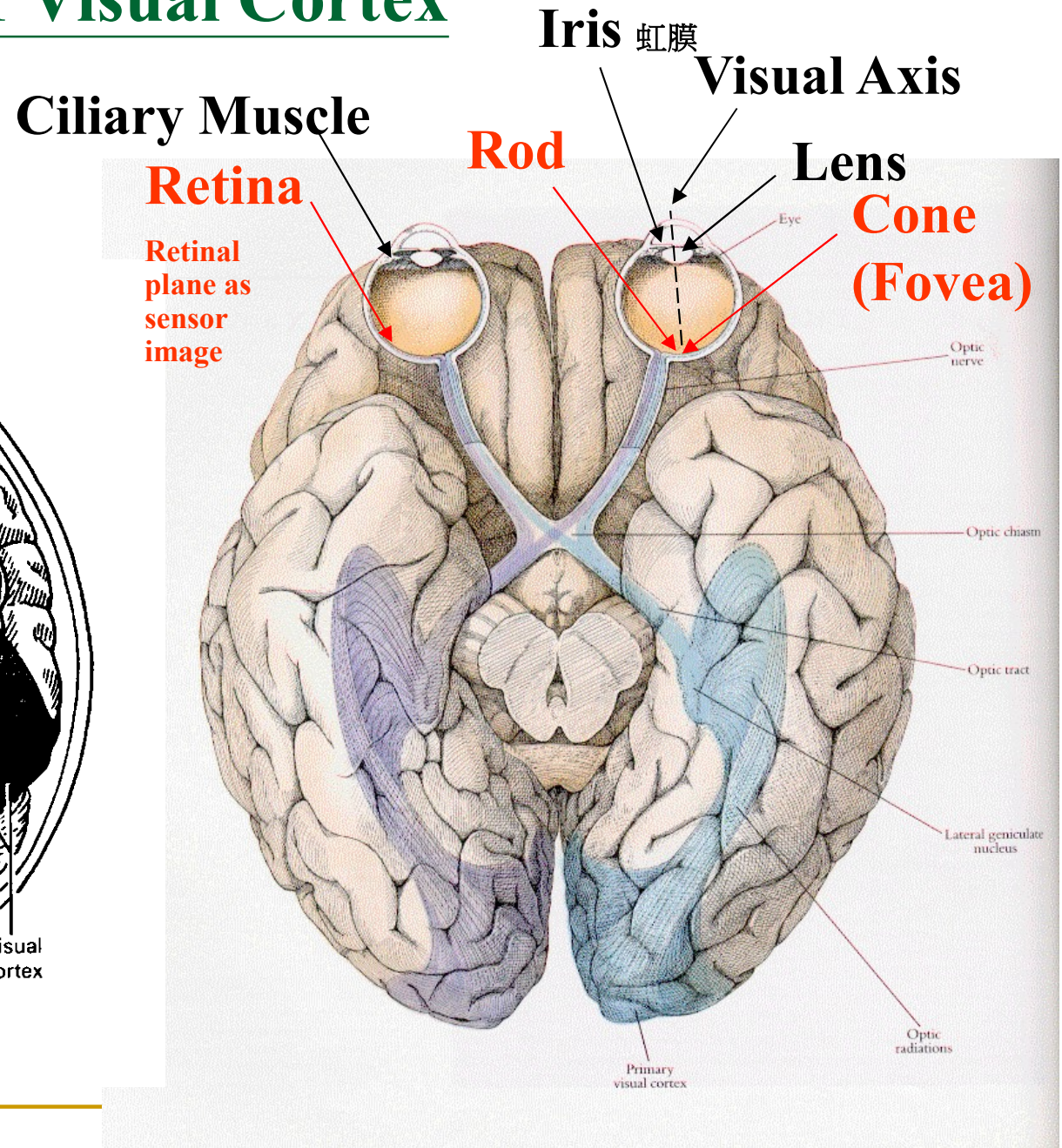
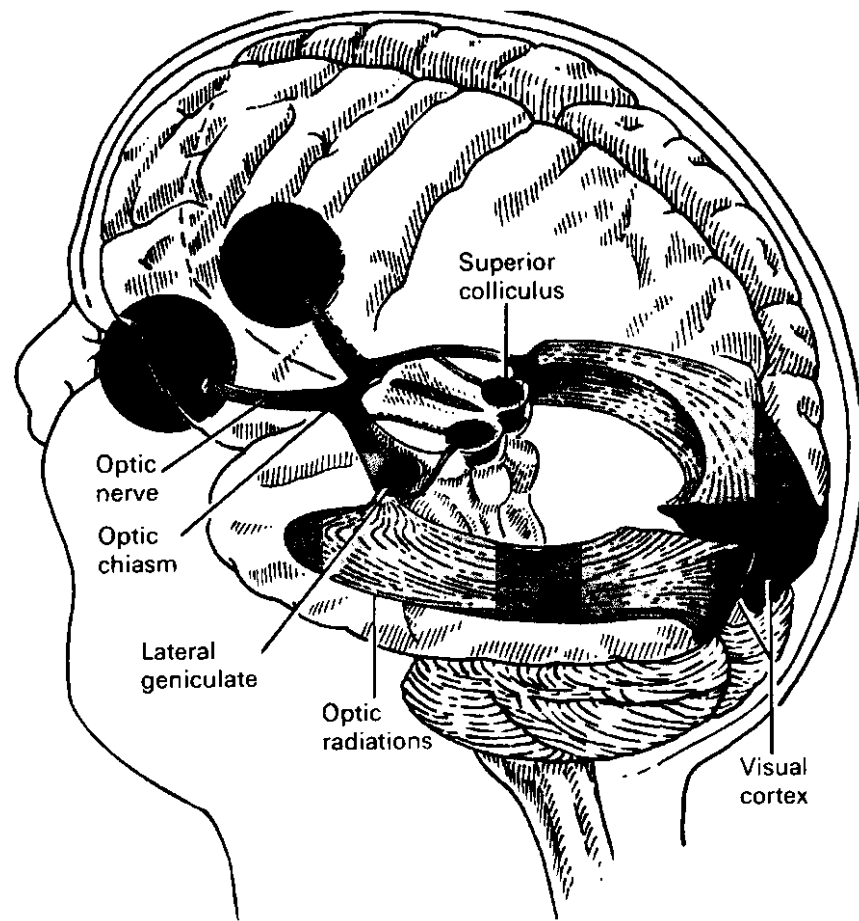

Image Format

教授：連震杰

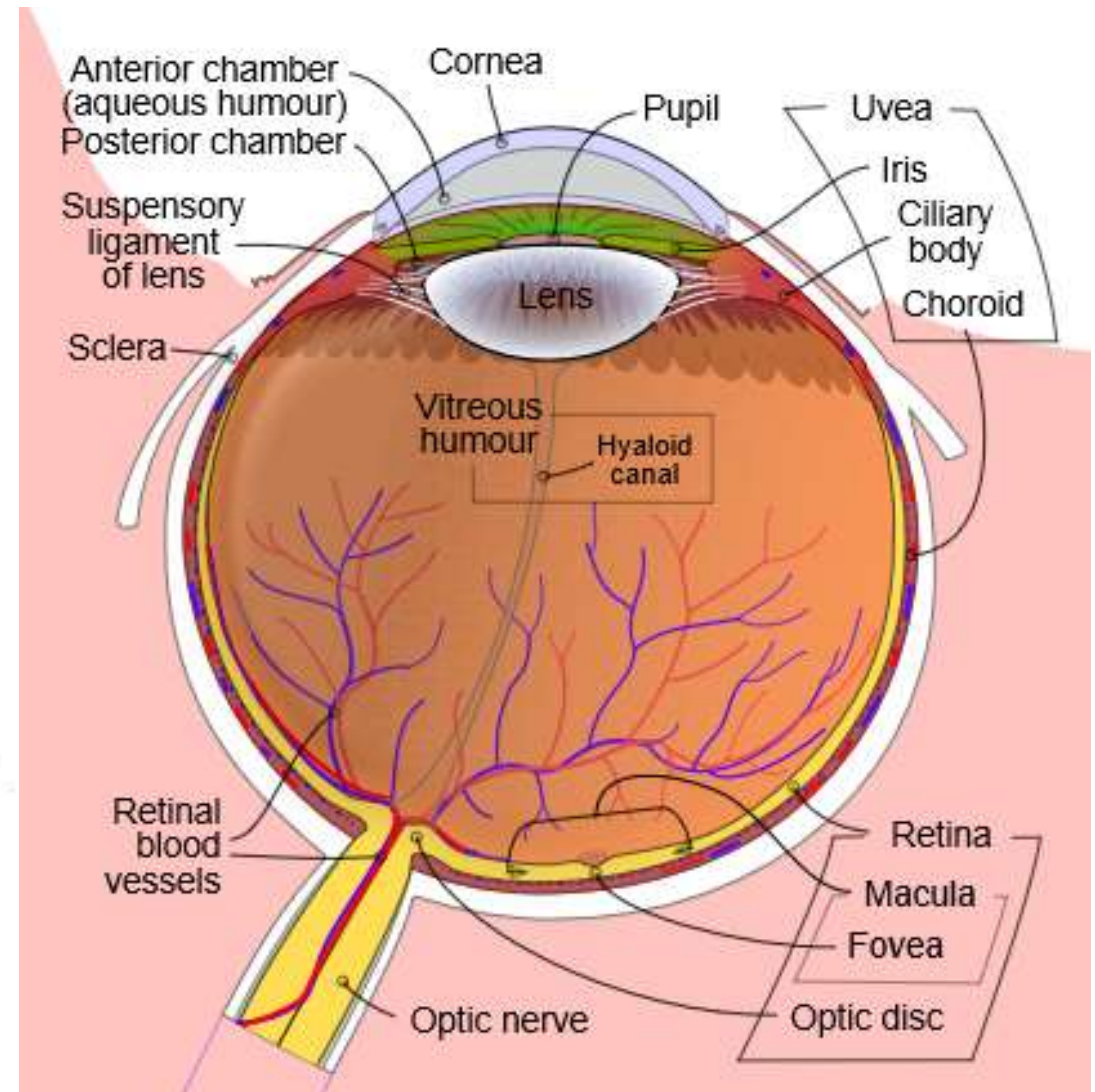
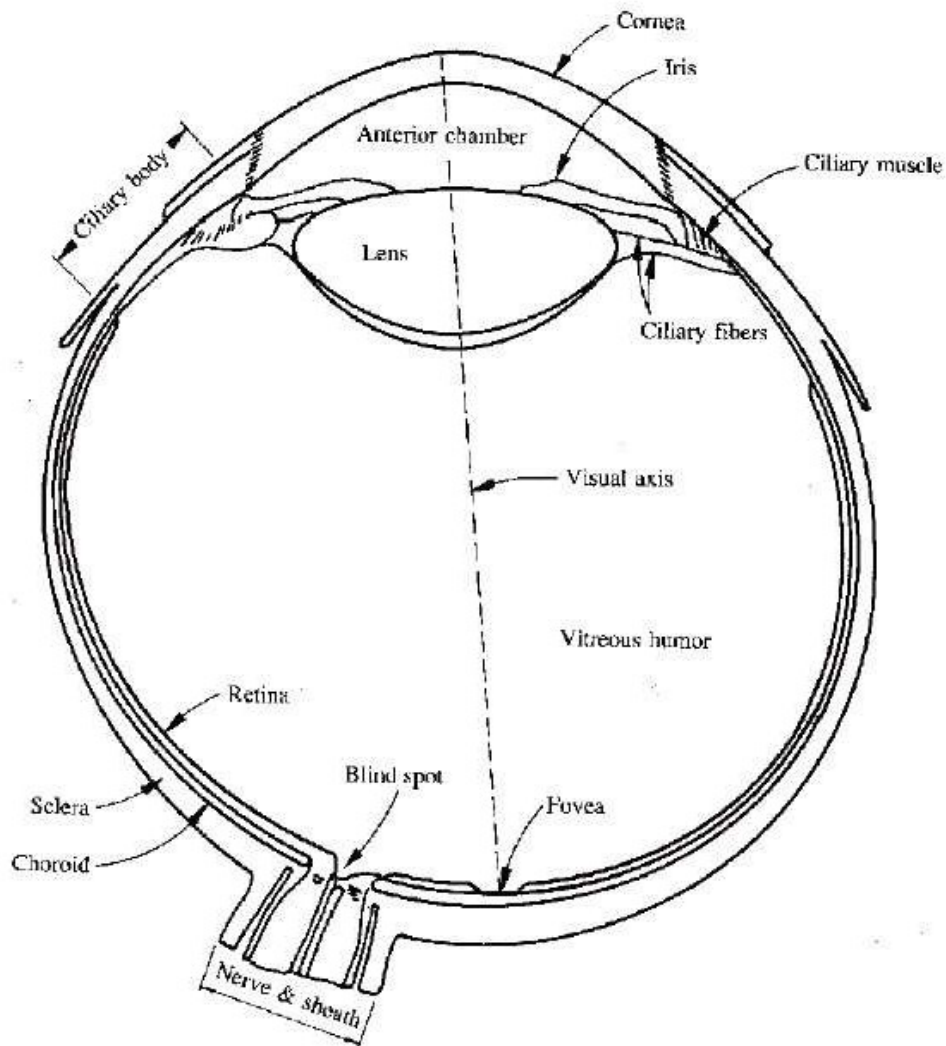
助教：

機器人實驗室

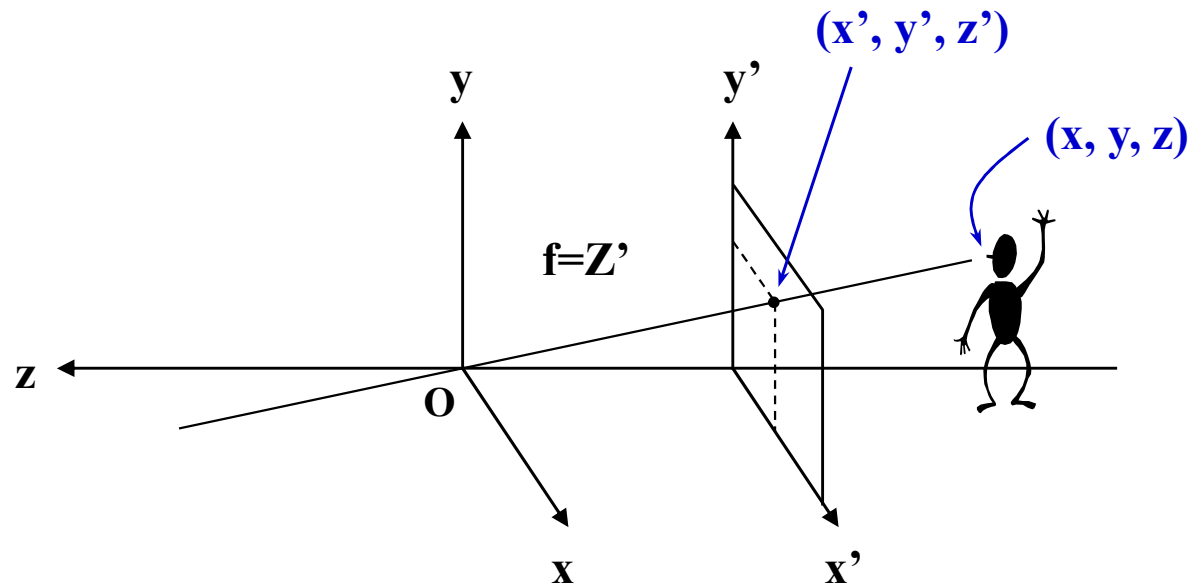
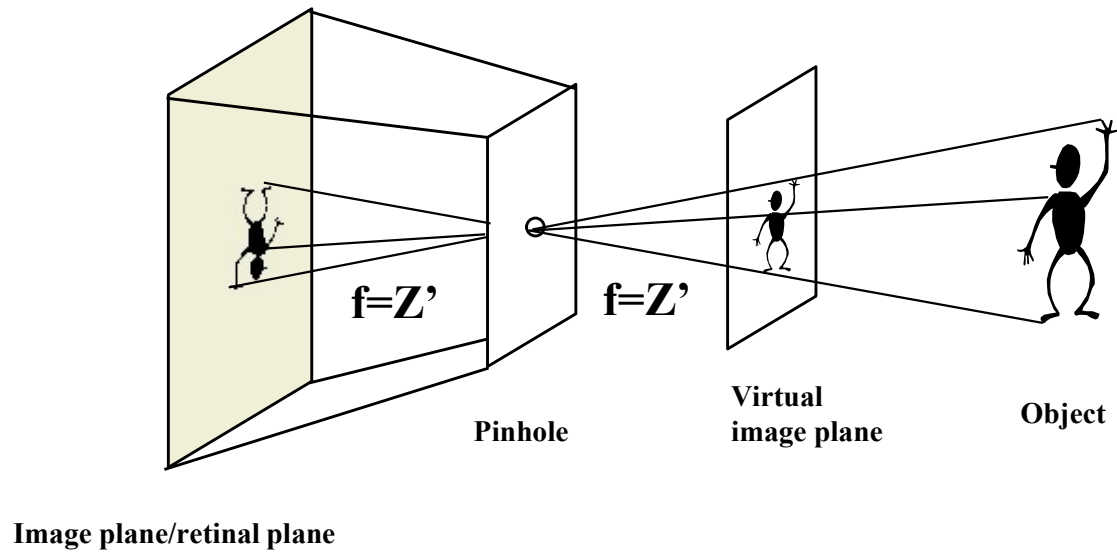
0.1 Visual Cortex



0.1 Cross Section of Human Eye



0.2 Pine Hole Camera Model: Perspective Projection



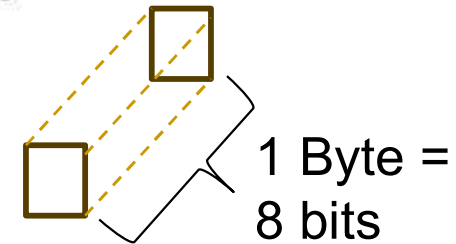
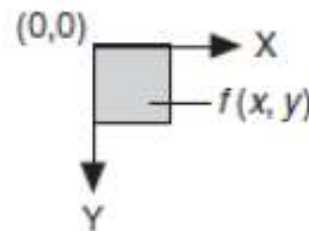
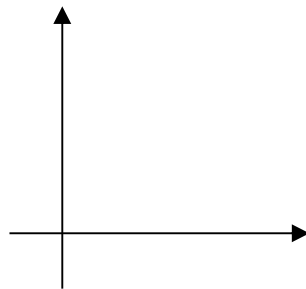
1.1 影像檔格式(Image Format) (1/3)

影像是以二維陣列來表示影像的光影強度大小。影像光強度方程式為

$$f(x, y) = I(x, y)$$

Intensity = grayvalue

x, y 分別代表二維陣列的行(column)與列(row)。



Pixel: Picture element
Voxel: Volume element
Dixel: Depth element

影像光強度方程式

數位影像處理式利用感測器，將影像轉換成離散的像素(pixels)，就是 $f(x, y)$ 所代表影像感測器(sensor)陣列的一個圖素(pixel)的強度。

1.1 影像檔格式(Image Format) (2/3)

影像與資料型態列表	
影像資料型態	單位 pixel 的 byte 數
1 bit , Line art	1/8 byte per pixel
1/8 byte per pixel	1/8 byte per pixel
8 bit , Grayscale 8 bits = 0~255	1 bytes per pixel
24 bit , RGB	3 bytes per pixel
32 Bit , CMYK	4 bytes per pixel
48 bit , RGB	6 bytes per pixel

CMYK:

C (Cyan) 青色 、 M (Magenta) 洋紅色 、 Y (Yellow) 黃色 、 K (Black) 黑色

1.1 影像檔格式(Image Format) (3/3)

影像檔格式	優點	缺點
1. BMP (Windows Bitmap)	專門用在Windows作業系統， 支援RGB全彩	無法壓縮 全彩影像，不適用於網路
2. GIF	網頁上最常用，可製作透明圖，交錯圖，動畫，提供 非破壞性壓縮(Lossless Compression) ，適用遠距傳遞，網頁設計	最多只能存256色 (8 bit, gray value??)
3. JPG/JPEG	高壓縮率 支援RGB全彩，灰階影像	破壞性壓縮， 易失真 不支援16色、256色、黑白圖
4. PNG (Portable Network Graphic)	非破壞性壓縮，結合GIF優點，JPEG 支援RGB全彩	無法儲存動畫
5. TGA	非破壞性壓縮，支援24位元(RGB)全彩、32位元全彩，用於影像合成，視訊合成，3D動畫製作的影像合成	需藉由專業影像處理軟體解讀
6. TIF (Tagged Image File Format)	跨平台， 非破壞性壓縮 ，支援RGB全彩、16色、256色、灰階、黑白影像	無
7. PCD	洗相片	柯達公司制定的相片光碟格式

1.2 點陣資料格式與取樣 (Raster Image and Sampling)

Raster Image (網格圖檔):

- 由掃描機、影像光碟、數位元元像機等設備或影像編輯軟體所產生之圖檔，為印前組頁檔案格式之一種。
- 其內容可分為灰階及彩色光點兩種，影像色彩輸出較真實；惟放大後會產生鋸齒狀現象。

Raster Graphics (點陣圖形):

(Dpi (dots per inch))

- 與位元影像圖形相同，是利用小點距來顯示出影像圖形，當點距越小，代表它的品質越高。

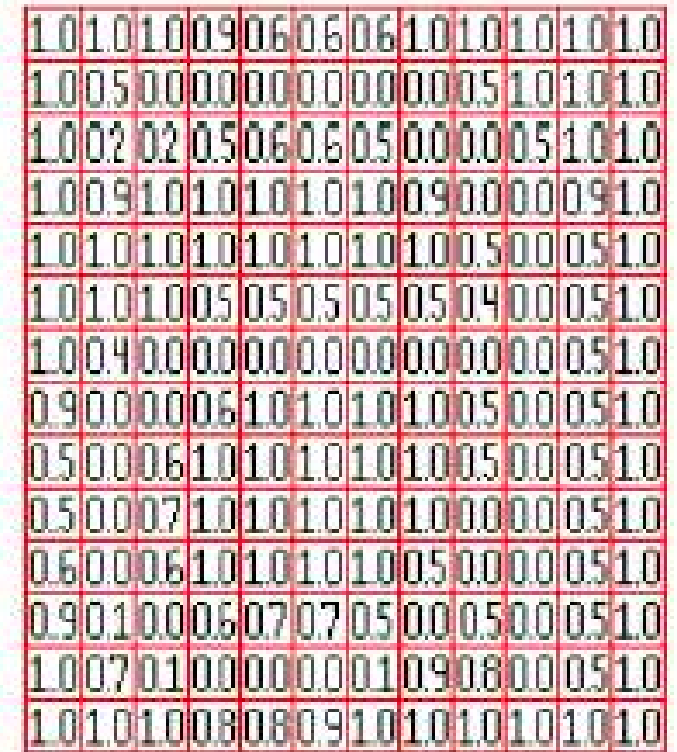
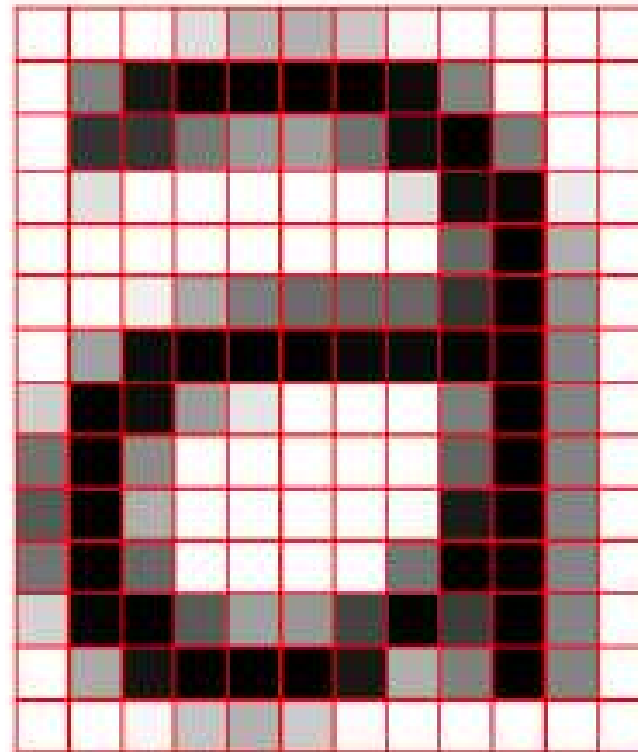
Gray value:

1.0 = 255 white,

0.0 = 0 black

Boundary blur problem

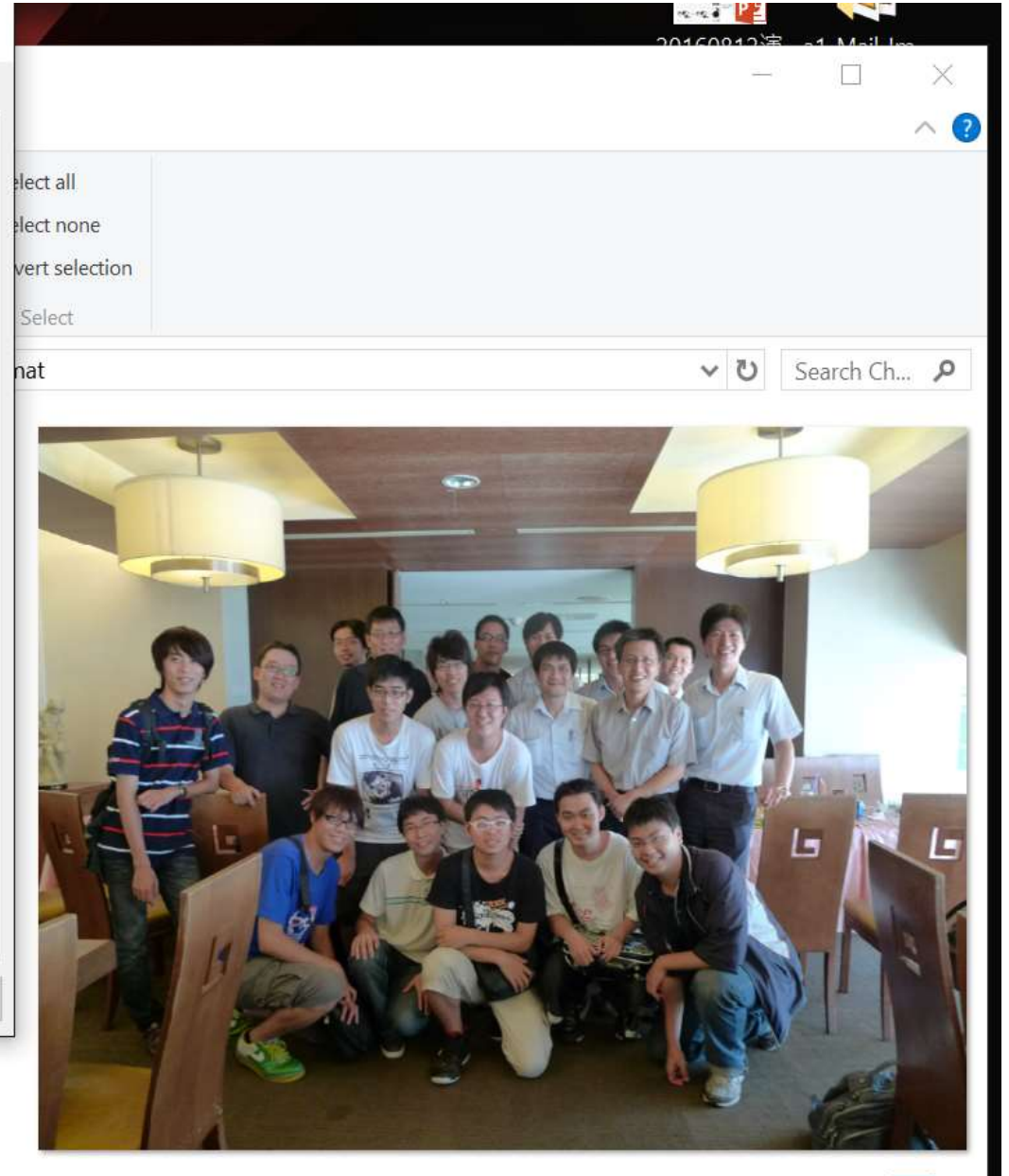
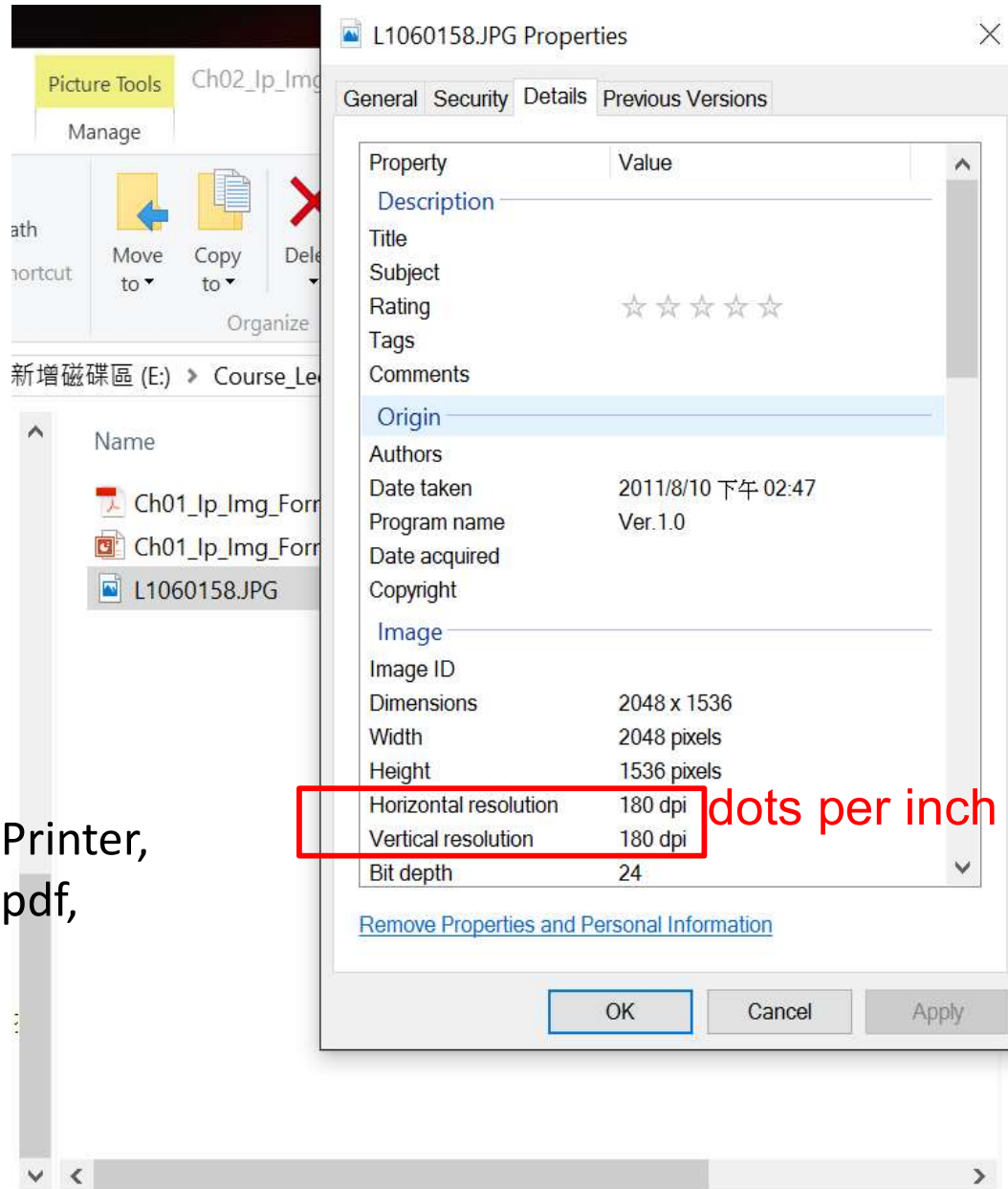
a



記憶體影像光強度

Line by edge detection → Not straight line
→ At least +/- 1 pixel offset

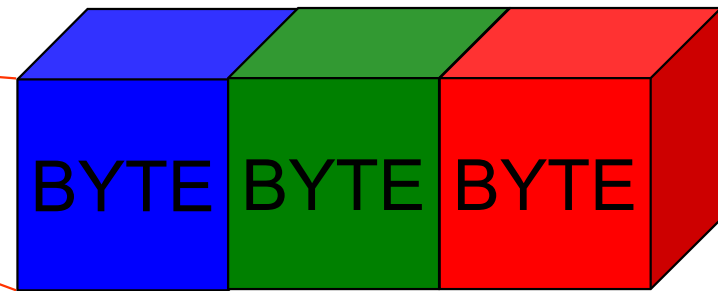
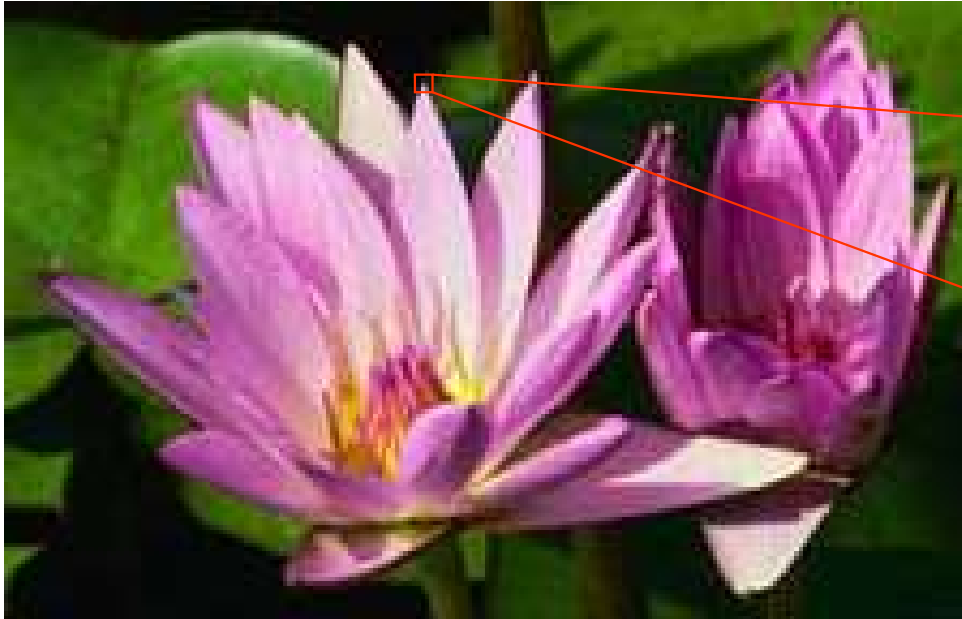
1.3 Image Property



2.0 What is BMP?

- The structure of BMP file was defined by Microsoft (BMP = Bitmap)
- Including **black-and-white** (1-bit per pixel), **gray value** (2 , 4 or 8 bits), **colorful** ($8*3=24$ or $8*4=32$ bits).
- Gray and black-and-white types have **palette**.
- We only deal with 24-bits BMP.

2.0 24-bit BMP file



1 BYTE=8 bits
=> 0 ~255



➤ Check ppt file color map

ASCII Code

1) BITMAPFILEHEADER (Bit Map File Header), 14 Bytes

3) **RGBQUAD** (RGB Quad, **palette**), $256 \cdot (3+1)$ Bytes

2.0 What's in a BMP file?

- 1) BITMAPFILEHEADER (Bit Map File Header) //14 bytes
- 2) BITMAPINFOHEADER (Bit Map Info Header) //40 bytes
- 3) **RGBQUAD * 2^N** (RGB Quad) // exp. N=8: 256*(3+1) bytes
(in our case)
- 4) Image Data Array

N (bits): How many bits stand for a pixel (black-and-white (1-bit per pixel), gray value (2 , 4 or 8 bits)).

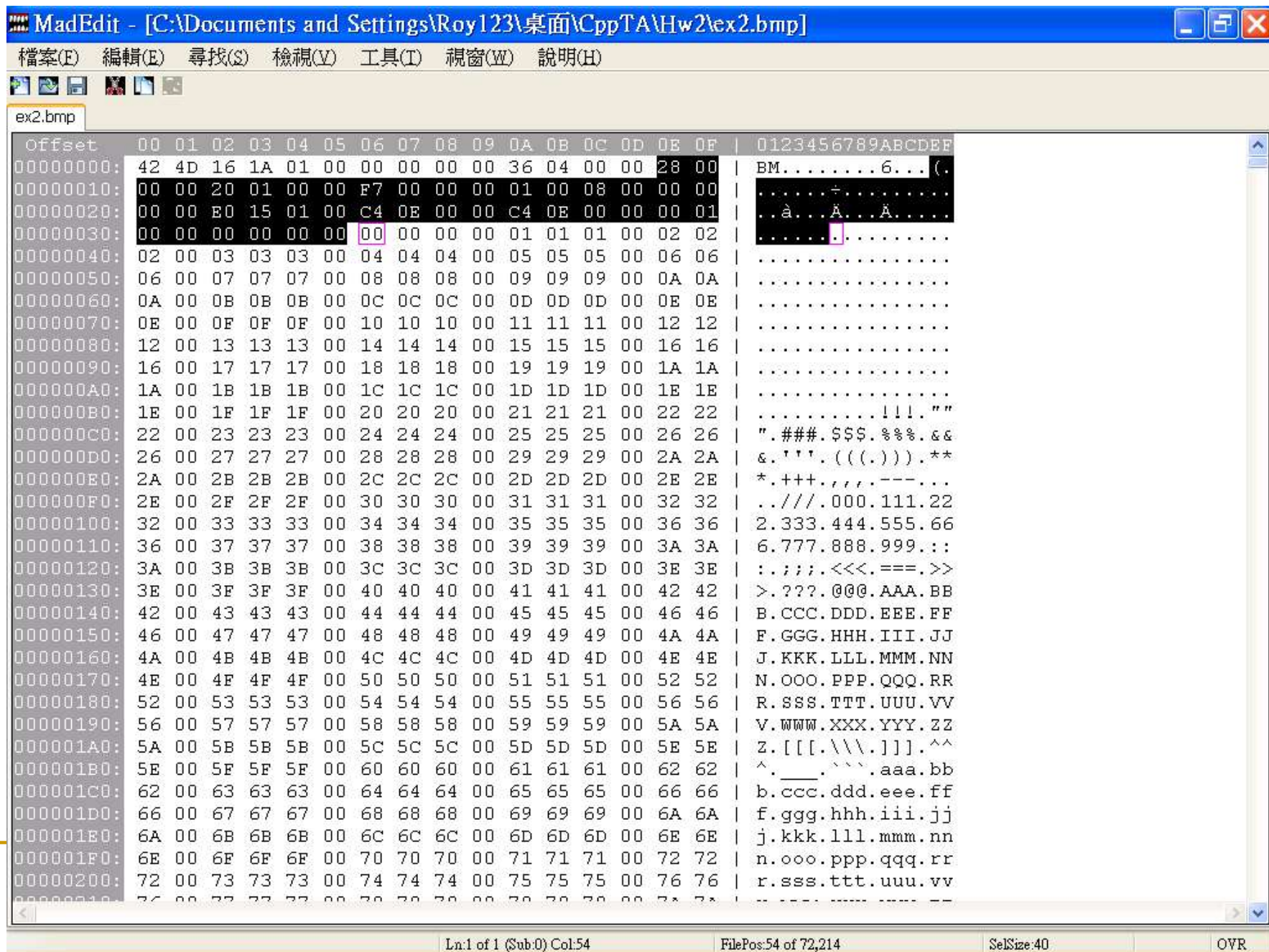
2.1 BITMAPFILEHEADER (1/2)

```
MadEdit - [C:\Documents and Settings\Roy123\桌面\CppTA\Hw2\ex2.bmp]
檔案(E) 編輯(E) 尋找(S) 檢視(V) 工具(T) 視窗(W) 說明(H)
ex2.bmp
Offset 00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F | 0123456789ABCDEF
00000000: 42 4D 16 1A 01 00 00 00 00 00 36 04 00 00 28 00 | BM.....6... (
00000010: 00 00 20 01 00 00 F7 00 00 00 01 00 08 00 00 00 | .....÷.....
00000020: 00 00 E0 15 01 00 C4 0E 00 00 C4 0E 00 00 00 01 | ..à...À...À....
00000030: 00 00 00 00 00 00 00 00 00 00 01 01 01 00 02 02 | .....
00000040: 02 00 03 03 03 00 04 04 04 00 05 05 05 00 06 06 | .....
00000050: 06 00 07 07 07 00 08 08 08 00 09 09 09 00 0A 0A | .....
00000060: 0A 00 0B 0B 0B 00 0C 0C 0C 00 0D 0D 0D 00 0E 0E | .....
00000070: 0E 00 0F 0F 0F 00 10 10 10 00 11 11 11 00 12 12 | .....
00000080: 12 00 13 13 13 00 14 14 14 00 15 15 15 00 16 16 | .....
00000090: 16 00 17 17 17 00 18 18 18 00 19 19 19 00 1A 1A | .....
000000A0: 1A 00 1B 1B 1B 00 1C 1C 1C 00 1D 1D 1D 00 1E 1E | .....
000000B0: 1E 00 1F 1F 1F 00 20 20 20 00 21 21 21 00 22 22 | .....! ! ! . "
000000C0: 22 00 23 23 23 00 24 24 24 00 25 25 25 00 26 26 | ".###.$$$.%%.&&
000000D0: 26 00 27 27 27 00 28 28 28 00 29 29 29 00 2A 2A | &.''.(((.)).**
000000E0: 2A 00 2B 2B 2B 00 2C 2C 2C 00 2D 2D 2D 00 2E 2E | *.+++.,,.,---...
000000F0: 2E 00 2F 2F 2F 00 30 30 30 00 31 31 31 00 32 32 | ..///.000.111.22
00000100: 32 00 33 33 33 00 34 34 34 00 35 35 35 00 36 36 | 2.333.444.555.66
00000110: 36 00 37 37 37 00 38 38 38 00 39 39 39 00 3A 3A | 6.777.888.999.:
00000120: 3A 00 3B 3B 3B 00 3C 3C 3C 00 3D 3D 3D 00 3E 3E | :.;;;.<<<==.>
00000130: 3E 00 3F 3F 3F 00 40 40 40 00 41 41 41 00 42 42 | >.???.@@@.AAA.BB
00000140: 42 00 43 43 43 00 44 44 44 00 45 45 45 00 46 46 | B.CCC.DDD.EEE.FF
00000150: 46 00 47 47 47 00 48 48 48 00 49 49 49 00 4A 4A | F.GGG.HHH.III.JJ
00000160: 4A 00 4B 4B 4B 00 4C 4C 4C 00 4D 4D 4D 00 4E 4E | J.KKK.LLL.MMM.NN
00000170: 4E 00 4F 4F 4F 00 50 50 50 00 51 51 51 00 52 52 | N.OOO.PPP.QQQ.RR
00000180: 52 00 53 53 53 00 54 54 54 00 55 55 55 00 56 56 | R.SSS.TTT.UUU.VV
00000190: 56 00 57 57 57 00 58 58 58 00 59 59 59 00 5A 5A | V.WWW.XXX.YYY.ZZ
000001A0: 5A 00 5B 5B 5B 00 5C 5C 5C 00 5D 5D 5D 00 5E 5E | Z.[[[.\ \ \.]]].^^
000001B0: 5E 00 5F 5F 5F 00 60 60 60 00 61 61 61 00 62 62 | ^.___.```.aaa.bb
000001C0: 62 00 63 63 63 00 64 64 64 00 65 65 65 00 66 66 | b.ccc.ddd.eee.ff
000001D0: 66 00 67 67 67 00 68 68 68 00 69 69 69 00 6A 6A | f.ggg.hhh.iii.jj
000001E0: 6A 00 6B 6B 6B 00 6C 6C 6C 00 6D 6D 6D 00 6E 6E | j.kkk.lll.mmm.nn
000001F0: 6E 00 6F 6F 6F 00 70 70 70 00 71 71 71 00 72 72 | n.ooo.ppp.qqq.rr
00000200: 72 00 73 73 73 00 74 74 74 00 75 75 75 00 76 76 | r.sss.ttt.uuu.vv
Ln:1 of 1 (Sub:0) Col:14 FilePos:14 of 72,214 SelSize:14 OVR
```

2.1 BITMAPFILEHEADER (2/2)

```
typedef struct tagBITMAPFILEHEADER {  
    WORD bfType;           //(2bytes)    // (14bytes)  
                             //          File type, in BMP  
                             //          case, it'll be  
                             //          'BM'(0x424D)  
    DWORD bfSize;          //(4bytes)    BMP file size  
    WORD bfReserved1;       //(2bytes)    Always 0  
    WORD bfReserved2;       //(2bytes)    Always 0  
    DWORD bfOffbytes;       //(4bytes)    Size of  
                             //          Headers (14+40  
                             //          bytes) + Palette,  
                             //          14+40 in our case  
} BITMAPFILEHEADER;
```

2.2 BITMAPINFOHEADER (1 / 2)



2.2 BITMAPINFOHEADER (2/2)

```
typedef struct tagBITMAPINFOHEADER{           //(40bytes)
    DWORD biSize;                //(4bytes) After Windows 3.X, it's always 40 (x28), which is
                                   // the structure size of BITMAPINFOHEADER
    LONG biWidth;                //(4bytes) The width of image
    LONG biHeight;               //(4bytes) The height of image
    WORD biPlanes;               //(2bytes) How many images in this file. For BMP, it's
                                   // always 1
    WORD biBitCount;             //(2bytes) How many bits stand for a pixel, 24 in our case
    DWORD biCompression;         //(4bytes) 0 is no compression, 1 is 8-bitRLE
                                   // compression, 2 is 4-bitRLE compression.
                                   // We only deal with no compression image.
    DWORD biSizeImage;           //(4bytes) The image size after compress. If no compression,
                                   // it could be 0 or image size
    LONG biXPelsPerMeter;        //(4bytes) horizontal dots per meter
    LONG biYPelsPerMeter;        //(4bytes) vertical dots per meter
    DWORD biClrUsed;             //(4bytes) How many colors used in palette, 0 for all colors
    DWORD biClrImportant;        //(4bytes) How many colors are important, 0 for all
} BITMAPINFOHEADER;
```

2.3 RGBQUAD (1/2)

MadEdit - [C:\Documents and Settings\Roy123\桌面\CppTA\Hw2\ex2.bmp]

檔案(E) 編輯(E) 尋找(S) 檢視(V) 工具(T) 視窗(W) 說明(H)

ex2.bmp

Offset	00	01	02	03	04	05	06	07	08	09	0A	0B	0C	0D	0E	0F	0123456789ABCDEF
00000000:	42	4D	16	1A	01	00	00	00	00	00	36	04	00	00	28	00	BM.....6... (
00000010:	00	00	20	01	00	00	F7	00	00	00	01	00	08	00	00	00	÷.....
00000020:	00	00	E0	15	01	00	C4	0E	00	00	C4	0E	00	00	00	01	..à...Ä...Ä....
00000030:	00	00	00	00	00	00	00	00	00	00	01	01	01	00	02	02	[.....]
00000040:	02	00	03	03	03	00	04	04	04	00	05	05	05	00	06	06	
00000050:	06	00	07	07	07	00	08	08	08	00	09	09	09	00	0A	0A	
00000060:	0A	00	0B	0B	0B	00	0C	0C	0C	00	0D	0D	0D	00	0E	0E	
00000070:	0E	00	0F	0F	0F	00	10	10	10	00	11	11	11	00	12	12	
00000080:	12	00	13	13	13	00	14	14	14	00	15	15	15	00	16	16	
00000090:	16	00	17	17	17	00	18	18	18	00	19	19	19	00	1A	1A	
000000A0:	1A	00	1B	1B	1B	00	1C	1C	1C	00	1D	1D	1D	00	1E	1E	
000000B0:	1E	00	1F	1F	1F	00	20	20	20	00	21	21	21	00	22	22	! ! ! . " "
000000C0:	22	00	23	23	23	00	24	24	24	00	25	25	25	00	26	26	".###.\$\$\$.%%.&&
000000D0:	26	00	27	27	27	00	28	28	28	00	29	29	29	00	2A	2A	&.''.(((.)).**
000000E0:	2A	00	2B	2B	2B	00	2C	2C	2C	00	2D	2D	2D	00	2E	2E	*.+++.,,.,--..
000000F0:	2E	00	2F	2F	2F	00	30	30	30	00	31	31	31	00	32	32	..///.000.111.22
00000100:	32	00	33	33	33	00	34	34	34	00	35	35	35	00	36	36	2.333.444.555.66
00000110:	36	00	37	37	37	00	38	38	38	00	39	39	39	00	3A	3A	6.777.888.999.::
00000120:	3A	00	3B	3B	3B	00	3C	3C	3C	00	3D	3D	3D	00	3E	3E	:.;;;<<<==.>
00000130:	3E	00	3F	3F	3F	00	40	40	40	00	41	41	41	00	42	42	>.???.@@@.AAA.BB
00000140:	42	00	43	43	43	00	44	44	44	00	45	45	45	00	46	46	B.CCC.DDD.EEE.FF
00000150:	46	00	47	47	47	00	48	48	48	00	49	49	49	00	4A	4A	F.GGG.HHH.III.JJ
00000160:	4A	00	4B	4B	4B	00	4C	4C	4C	00	4D	4D	4D	00	4E	4E	J.KKK.LLL.MMM.NN
00000170:	4E	00	4F	4F	4F	00	50	50	50	00	51	51	51	00	52	52	N.OOO.PPP.QQQ.RR
00000180:	52	00	53	53	53	00	54	54	54	00	55	55	55	00	56	56	R.SSS.TTT.UUU.VV
00000190:	56	00	57	57	57	00	58	58	58	00	59	59	59	00	5A	5A	V.WWW.XXX.YYY.ZZ
000001A0:	5A	00	5B	5B	5B	00	5C	5C	5C	00	5D	5D	5D	00	5E	5E	Z.[[[.\.\.]].^^
000001B0:	5E	00	5F	5F	5F	00	60	60	60	00	61	61	61	00	62	62	^.___.```.aaa.bb
000001C0:	62	00	63	63	63	00	64	64	64	00	65	65	65	00	66	66	b.ccc.ddd.eee.ff
000001D0:	66	00	67	67	67	00	68	68	68	00	69	69	69	00	6A	6A	f.ggg.hhh.iii.jj
000001E0:	6A	00	6B	6B	6B	00	6C	6C	6C	00	6D	6D	6D	00	6E	6E	j.kkk.lll.mmm.nn
000001F0:	6E	00	6F	6F	6F	00	70	70	70	00	71	71	71	00	72	72	n.ooo.ppp.qqq.rr
00000200:	72	00	73	73	73	00	74	74	74	00	75	75	75	00	76	76	r.sss.ttt.uuu.vv
00000210:	76	00	77	77	77	00	78	78	78	00	79	79	79	00	7A	7A	v.www.xxx.yyy.zz
00000220:	7A	00	7B	7B	7B	00	7C	7C	7C	00	7D	7D	7D	00	7E	7E	z.{{{. .}}}.~~
00000230:	7E	00	7F	7F	7F	00	80	80	80	00	81	81	81	00	82	82	~. .ooo.ooo.oo

2.3 RGBQUAD (2/2)

```
typedef struct tagRGBQUAD{                                //(4bytes)
    BYTE rgbBlue;          //(1bytes)    blue channel
    BYTE rgbGreen;         //(1bytes)    green channel
    BYTE rgbRed;           //(1bytes)    red channel
    BYTE rgbReserved;      //(1bytes)    must be 0
} RGBQUAD;
```

// Palette - $256 \times (3+1)$ bytes

LUT

2.4 Image Data Array (1/4)

```
MadEdit - [C:\Documents and Settings\Roy123\桌面\圖片1.bmp]
檔案(F) 編輯(E) 尋找(S) 檢視(V) 工具(T) 視窗(W) 說明(H)
圖片1.bmp
Offset 00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F | 0123456789ABCDEF
00000000: 42 4D 36 C0 0D 00 00 00 00 00 36 00 00 00 28 00 | BM6A.....6... (.
00000010: 00 00 AA 02 00 00 B8 01 00 00 01 00 18 00 00 00 | ...a.....
00000020: 00 00 00 C0 0D 00 12 17 00 00 12 17 00 00 00 00 | ...A.....
00000030: 00 00 00 00 00 00 FF FF FF FF FF FF FF FF FF FF | .....yyyyyyyyyy
00000040: FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF | yyyyyyyyyyyyyyyy
00000050: FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF | yyyyyyyyyyyyyyyy
00000060: FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF | yyyyyyyyyyyyyyyy
00000070: FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF | yyyyyyyyyyyyyyyy
00000080: FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF | yyyyyyyyyyyyyyyy
00000090: FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF | yyyyyyyyyyyyyyyy
000000A0: FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF | yyyyyyyyyyyyyyyy
000000B0: FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF | yyyyyyyyyyyyyyyy
000000C0: FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF | yyyyyyyyyyyyyyyy
000000D0: FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF | yyyyyyyyyyyyyyyy
000000E0: FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF | yyyyyyyyyyyyyyyy
000000F0: FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF | yyyyyyyyyyyyyyyy
00000100: FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF | yyyyyyyyyyyyyyyy
00000110: FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF | yyyyyyyyyyyyyyyy
00000120: FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF | yyyyyyyyyyyyyyyy
00000130: FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF | yyyyyyyyyyyyyyyy
00000140: FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF | yyyyyyyyyyyyyyyy
00000150: FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF | yyyyyyyyyyyyyyyy
00000160: FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF | yyyyyyyyyyyyyyyy
00000170: FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF | yyyyyyyyyyyyyyyy
00000180: FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF | yyyyyyyyyyyyyyyy
00000190: FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF | yyyyyyyyyyyyyyyy
000001A0: FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF | yyyyyyyyyyyyyyyy
000001B0: FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF | yyyyyyyyyyyyyyyy
000001C0: FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF | yyyyyyyyyyyyyyyy
000001D0: FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF | yyyyyyyyyyyyyyyy
000001E0: FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF | yyyyyyyyyyyyyyyy
000001F0: FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF | yyyyyyyyyyyyyyyy
00000200: FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF | yyyyyyyyyyyyyyyy
00000210: FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF | yyyyyyyyyyyyyyyy
00000220: FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF | yyyyyyyyyyyyyyyy
00000230: FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF | yyyyyyyyyyyyyyyy
00000240: FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF | yyyyyyyyyyyyyyyy
00000250: FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF | yyyyyyyyyyyyyyyy
00000260: FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF | yyyyyyyyyyyyyyyy
00000270: FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF | yyyyyyyyyyyyyyyy
00000280: FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF | yyyyyyyyyyyyyyyy
```

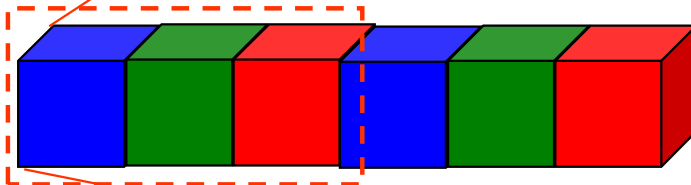
2.4 Image Data Array (2/4)

682

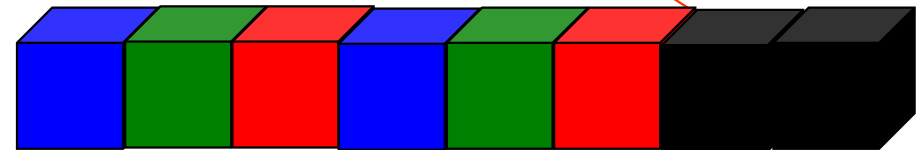
440



1 Pixel



...



Width Vs. Step

$$682 * 3 = 2046 \text{ (BYTE)}$$

2.4 Image Data Array (3/4)

MadEdit - [C:\Documents and Settings\Roy123\桌面\圖片1.bmp]

檔案(F) 編輯(E) 尋找(S) 檢視(V) 工具(T) 視窗(W) 說明(H)

圖片1.bmp

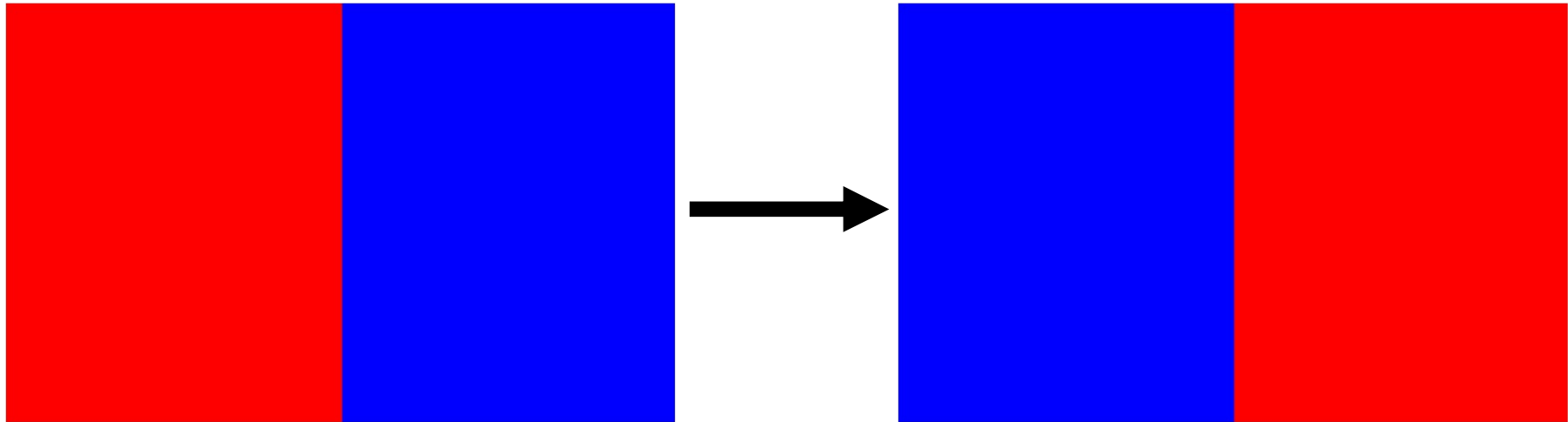
Offset	00	01	02	03	04	05	06	07	08	09	0A	0B	0C	0D	0E	0F	0123456789ABCDEF
00000690:	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	YYYYYYYYYYYYYYYY
000006A0:	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	YYYYYYYYYYYYYYYY
000006B0:	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	YYYYYYYYYYYYYYYY
000006C0:	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	YYYYYYYYYYYYYYYY
000006D0:	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	YYYYYYYYYYYYYYYY
000006E0:	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	YYYYYYYYYYYYYYYY
000006F0:	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	YYYYYYYYYYYYYYYY
00000700:	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	YYYYYYYYYYYYYYYY
00000710:	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	YYYYYYYYYYYYYYYY
00000720:	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	YYYYYYYYYYYYYYYY
00000730:	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	YYYYYYYYYYYYYYYY
00000740:	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	YYYYYYYYYYYYYYYY
00000750:	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	YYYYYYYYYYYYYYYY
00000760:	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	YYYYYYYYYYYYYYYY
00000770:	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	YYYYYYYYYYYYYYYY
00000780:	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	YYYYYYYYYYYYYYYY
00000790:	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	YYYYYYYYYYYYYYYY
000007A0:	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	YYYYYYYYYYYYYYYY
000007B0:	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	YYYYYYYYYYYYYYYY
000007C0:	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	YYYYYYYYYYYYYYYY
000007D0:	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	YYYYYYYYYYYYYYYY
000007E0:	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	YYYYYYYYYYYYYYYY
000007F0:	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	YYYYYYYYYYYYYYYY
00000800:	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	YYYYYYYYYYYYYYYY
00000810:	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	YYYYYYYYYYYYYYYY
00000820:	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	YYYYYYYYYYYYYYYY
00000830:	FF	FF	FF	FF	00	00	FF	FF	FF	00	11	09	00	10	08	00	YYYY..YY.....
00000840:	0E	07	00	0B	06	00	09	03	00	07	01	01	06	01	03	06	
00000850:	01	06	07	02	09	09	04	0C	0B	07	10	0C	09	10	0E	0A	
00000860:	11	0E	0A	10	0D	09	0E	0B	07	0B	0A	06	08	08	04	04	
00000870:	07	03	02	07	02	01	08	03	02	09	04	02	09	04	02	09	
00000880:	04	02	0A	05	02	09	04	02	09	04	02	09	04	03	09	04	
00000890:	04	09	04	06	08	04	08	08	04	0A	08	04	0A	08	04	0B	
000008A0:	08	04	0C	08	04	0C	07	04	0D	07	04	0D	07	04	0D	07	
000008B0:	04	0C	08	04	0A	08	04	09	08	04	08	08	04	06	08	04	
000008C0:	06	09	04	05	09	04	05	09	05	06	08	05	06	07	06	07	
000008D0:	07	07	07	07	07	07	07	07	07	07	07	07	07	07	07	07	
000008E0:	07	07	07	07	07	07	07	07	07	07	07	07	07	07	07	07	
000008F0:	07	07	07	07	07	07	07	07	07	07	07	07	07	07	07	07	
00000900:	07	07	07	07	07	07	07	07	07	07	07	07	07	07	07	07	
00000910:	07	07	07	07	07	07	07	07	07	07	07	07	07	07	07	07	

Ln:1 of 1 (Sub:0) Col:2,102 FilePos:2,102 of 901,174

2.4 Image Data Array (4/4)

- Image data array is a huge array that record all pixel values of image.
- Each row should be **a multiple of 4**.
- If the width of image wasn't a multiple of 4, we should **fill up with zeros**.
 - ex: width = 13 → fill up with 3 zeros, then the new width = 16
- The order of image is **upper-side-down**, the last row in image is the first row in file. It is because of displayer - **Scanning**.

2.5 Others: Mirror

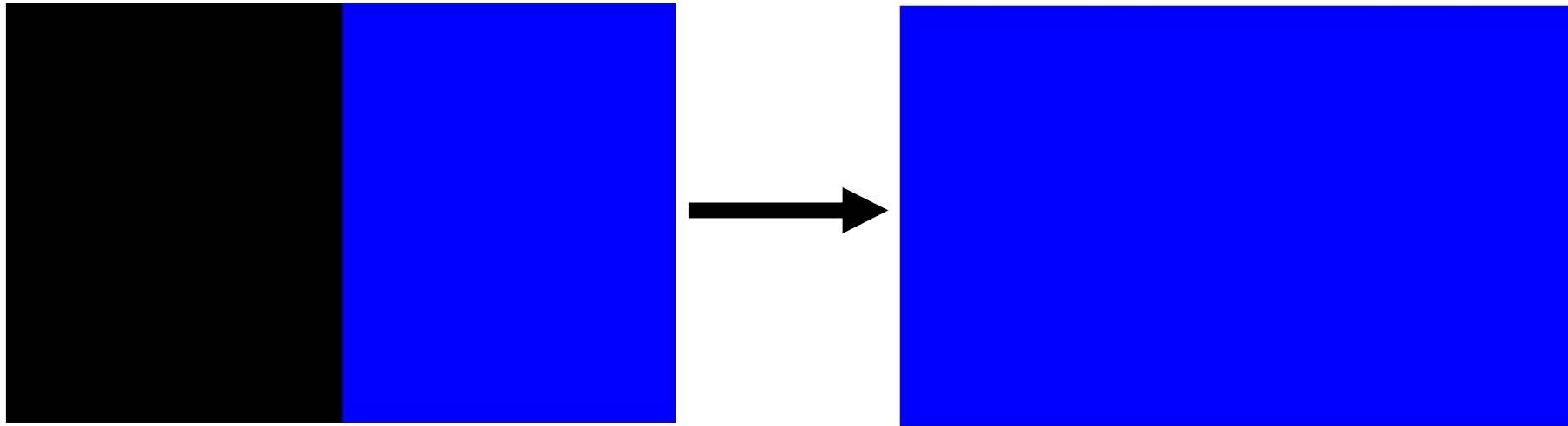


255,0,0	255,0,0	0,0,255	0,0,255
255,0,0	255,0,0	0,0,255	0,0,255
255,0,0	255,0,0	0,0,255	0,0,255



0,0,255	0,0,255	255,0,0	255,0,0
0,0,255	0,0,255	255,0,0	255,0,0
0,0,255	0,0,255	255,0,0	255,0,0

2.5 Others: Color change



0,0,0	0,0,0	0,0,255	0,0,255
0,0,0	0,0,0	0,0,255	0,0,255
0,0,0	0,0,0	0,0,255	0,0,255



0,0,255	0,0,255	0,0,255	0,0,255
0,0,255	0,0,255	0,0,255	0,0,255
0,0,255	0,0,255	0,0,255	0,0,255

2.5 Others: Embossment (1/3)



- Converting the input image to the gray level image.
- Convolute with 2 sobel mask, vertical and horizontal → $[-xx, +xx]$
- Then normalize to $[0, 255]$

2.5 Others: Embossment (2/3)

Vertical

-1	0	1
-2	0	2
-1	0	1

Horizontal

-1	-2	-1
0	0	0
1	2	1

Sobel Mask or Filter

- You may leave the boundary of image without dealing with it.
- New pixel value was calculated by neighboring pixels.

2.5 Others: Embossment (3/3)

Filter		Original image																			
<table><tr><td>-1</td><td>0</td><td>1</td></tr><tr><td>-2</td><td>0</td><td>2</td></tr><tr><td>-1</td><td>0</td><td>1</td></tr></table>	-1	0	1	-2	0	2	-1	0	1	*	<table><tr><td>(x-1, y-1)</td><td>(x, y-1)</td><td>(x+1, y-1)</td></tr><tr><td>(x-1, y)</td><td>(x, y)</td><td>(x+1, y)</td></tr><tr><td>(x-1, y+1)</td><td>(x, y+1)</td><td>(x+1, y+1)</td></tr></table>	(x-1, y-1)	(x, y-1)	(x+1, y-1)	(x-1, y)	(x, y)	(x+1, y)	(x-1, y+1)	(x, y+1)	(x+1, y+1)	+
-1	0	1																			
-2	0	2																			
-1	0	1																			
(x-1, y-1)	(x, y-1)	(x+1, y-1)																			
(x-1, y)	(x, y)	(x+1, y)																			
(x-1, y+1)	(x, y+1)	(x+1, y+1)																			
<table><tr><td>-1</td><td>-2</td><td>-1</td></tr><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>1</td><td>2</td><td>1</td></tr></table>	-1	-2	-1	0	0	0	1	2	1	*	<table><tr><td>(x-1, y-1)</td><td>(x, y-1)</td><td>(x+1, y-1)</td></tr><tr><td>(x-1, y)</td><td>(x, y)</td><td>(x+1, y)</td></tr><tr><td>(x-1, y+1)</td><td>(x, y+1)</td><td>(x+1, y+1)</td></tr></table>	(x-1, y-1)	(x, y-1)	(x+1, y-1)	(x-1, y)	(x, y)	(x+1, y)	(x-1, y+1)	(x, y+1)	(x+1, y+1)	=
-1	-2	-1																			
0	0	0																			
1	2	1																			
(x-1, y-1)	(x, y-1)	(x+1, y-1)																			
(x-1, y)	(x, y)	(x+1, y)																			
(x-1, y+1)	(x, y+1)	(x+1, y+1)																			

New value of central pixel

- New value normalizes to $\rightarrow [0, 255]$

2.5.1 Step 1 - Design Network Structure / Topology: Feature Extraction Maps by Convolution (1/3)

CNN – Convolution

1 : ink
0 : no ink

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image

Weight w ?
Bias b ?

Those are the network parameters to be learned.

1	-1	-1
-1	1	-1
-1	-1	1

Filter 1
Matrix

Some features

-1	1	-1
-1	1	-1
-1	1	-1

Filter 2
Matrix

⋮

Filter – Feature
description/extraction

Property 1

Each filter detects a small pattern (3 x 3).

J: Here, filter parameters are the weights and bias

2.5.1 Step 1 - Convolutional neural networks for isolated character recognition

Convolutional layer

1 _{x1}	1 _{x0}	1 _{x1}	0	0
0 _{x0}	1 _{x1}	1 _{x0}	1	0
0 _{x1}	0 _{x0}	1 _{x1}	1	1
0	0	1	1	0
0	1	1	0	0

Image

4		

Convolved
Feature

$$\begin{aligned} &1 \times 1 + 1 \times 0 + 1 \times 1 + \\ &0 \times 0 + 1 \times 1 + 1 \times 0 + \\ &0 \times 1 + 0 \times 0 + 1 \times 1 \\ &= 4 \end{aligned}$$

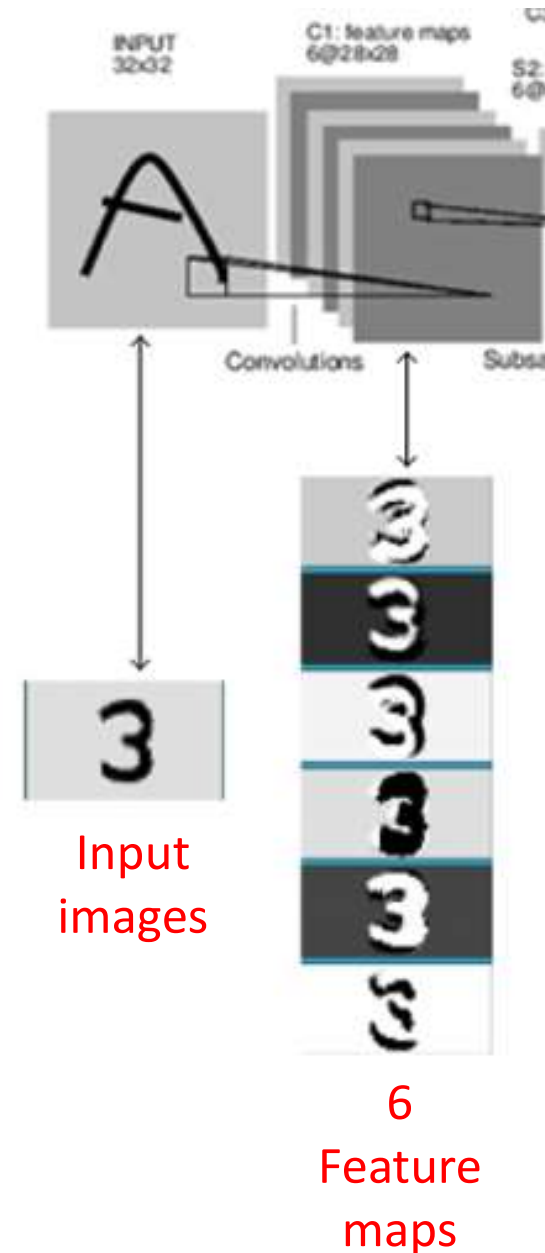
1	1	1	0	0
0	1	1	1	0
0 _{x1}	0 _{x0}	1 _{x1}	1	1
0 _{x0}	0 _{x1}	1 _{x0}	1	0
0 _{x1}	1 _{x0}	1 _{x1}	0	0

Image

4	3	4
2	4	3
2		

Convolved
Feature

$$\begin{aligned} &0 \times 1 + 0 \times 0 + 1 \times 1 + \\ &0 \times 0 + 0 \times 1 + 1 \times 0 + \\ &0 \times 1 + 1 \times 0 + 1 \times 1 \\ &= 2 \end{aligned}$$



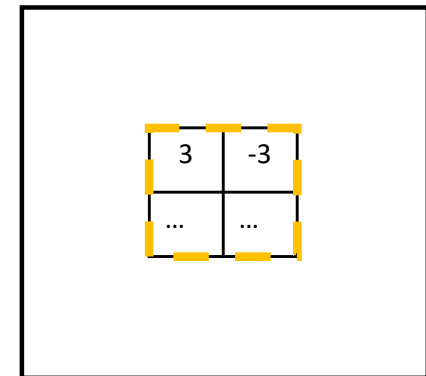
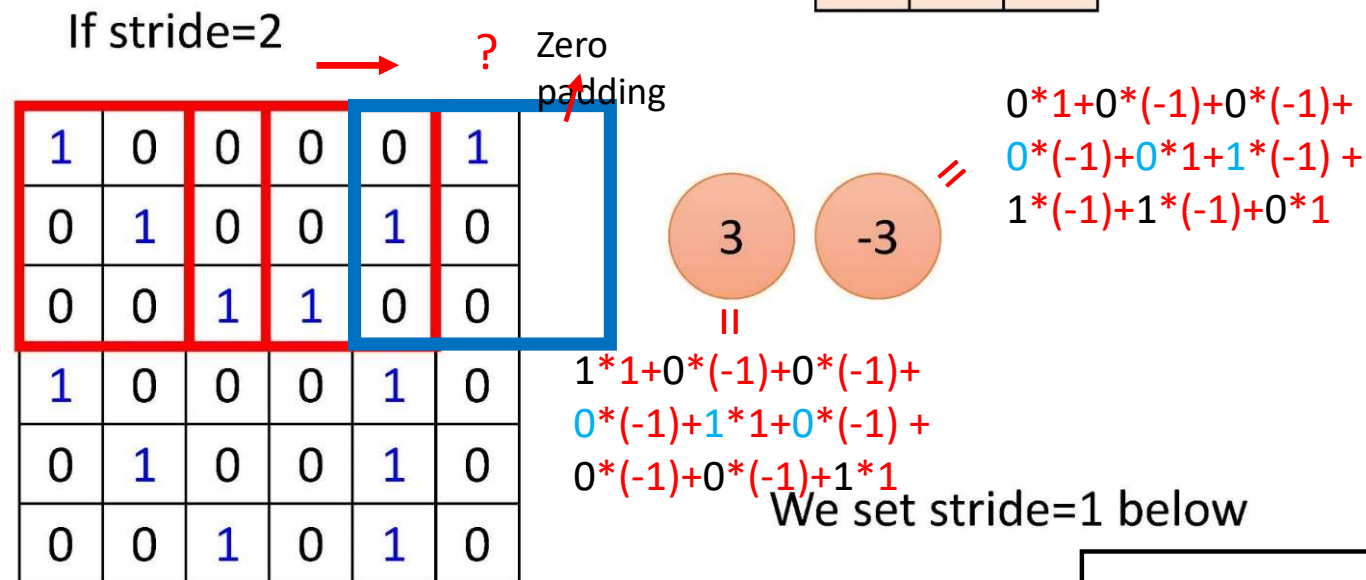
How convolutional layer works and its result

2.5.1 Step 1 - Design Network Structure / Topology: Feature Extraction Maps by Convolution (2/3)

CNN – Convolution

1	-1	-1
-1	1	-1
-1	-1	1

Filter 1



2.5.1 Step 1 - Design Network Structure / Topology: Feature Extraction Maps by Convolution (3/3)

CNN – Convolution

